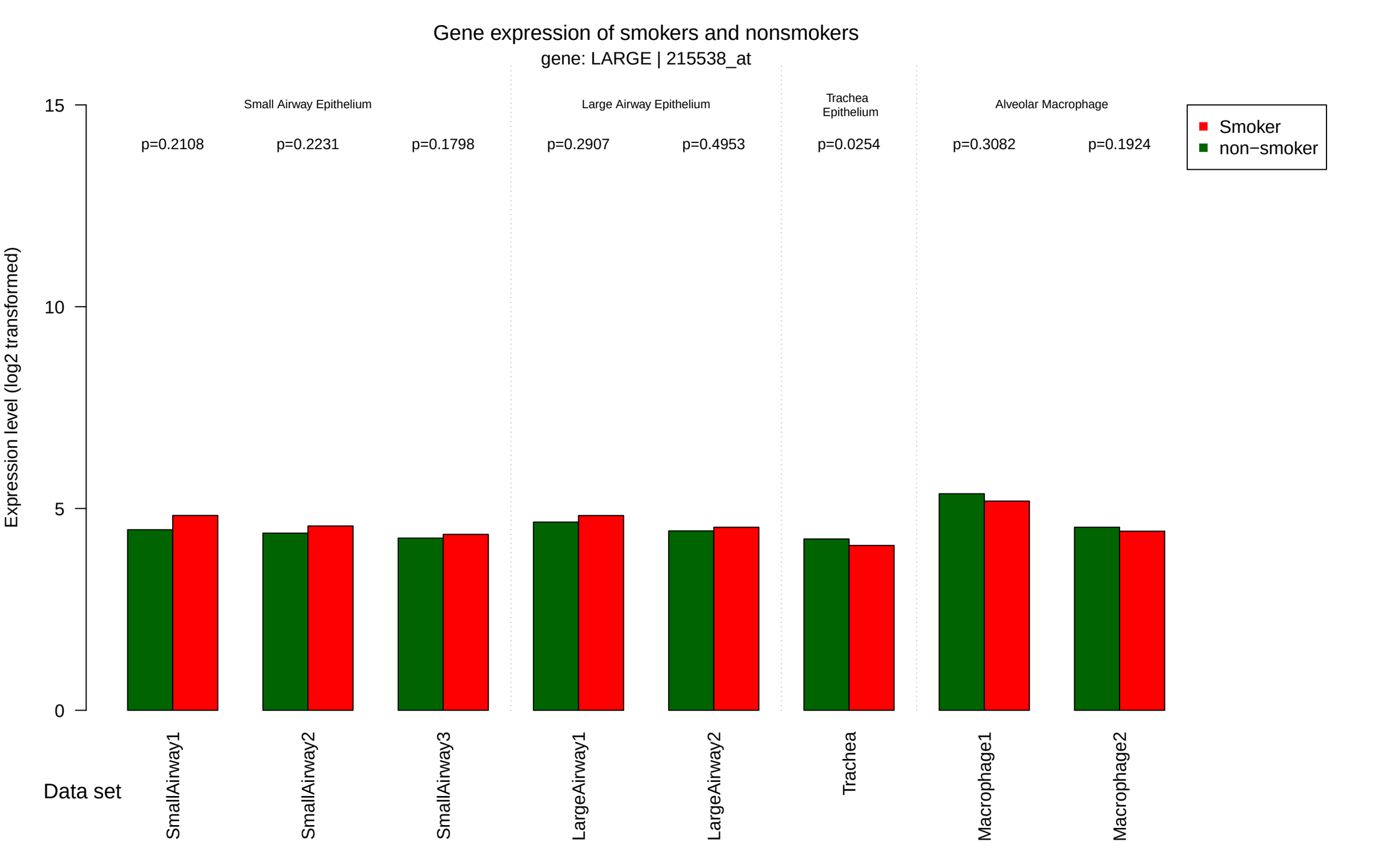
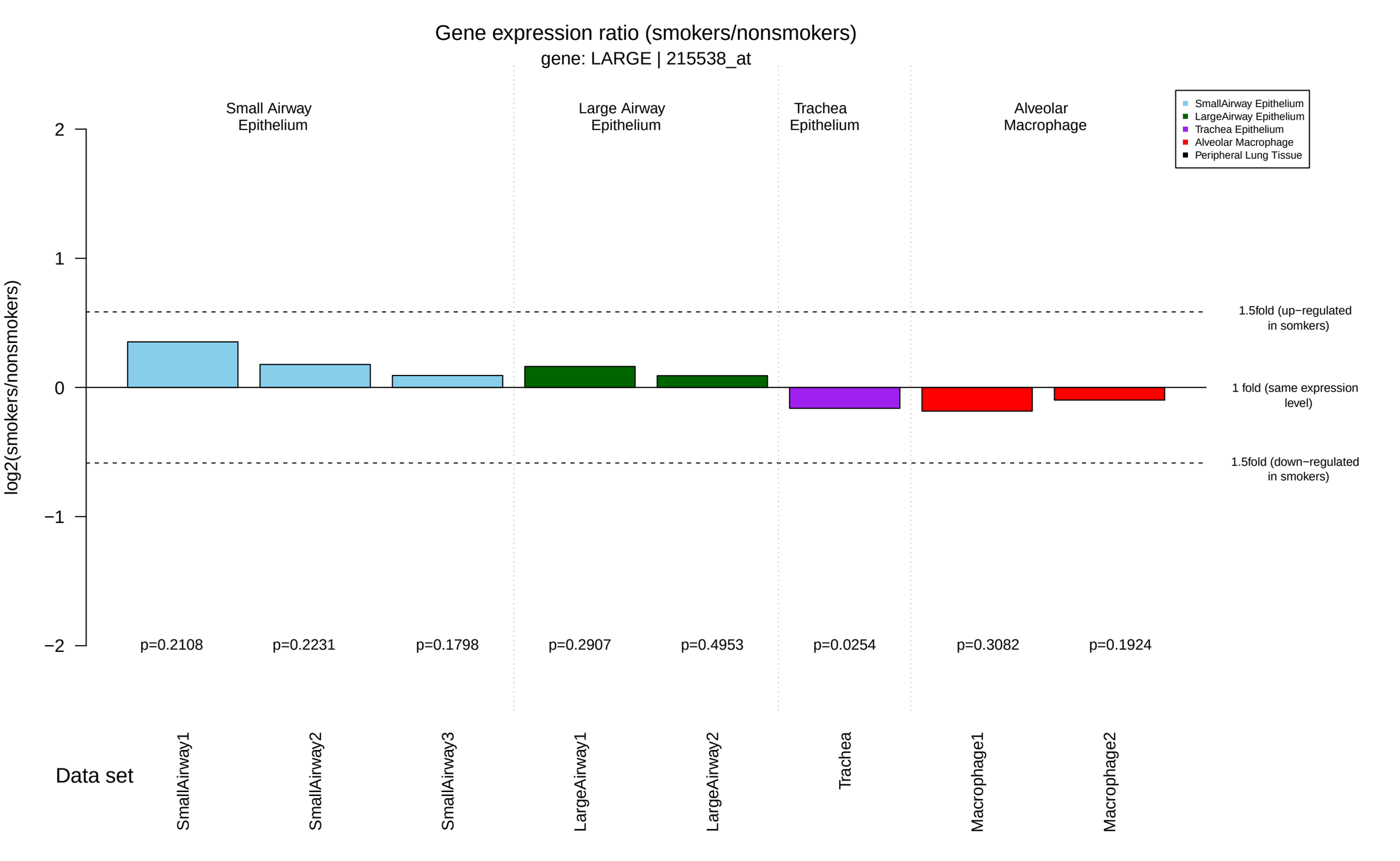
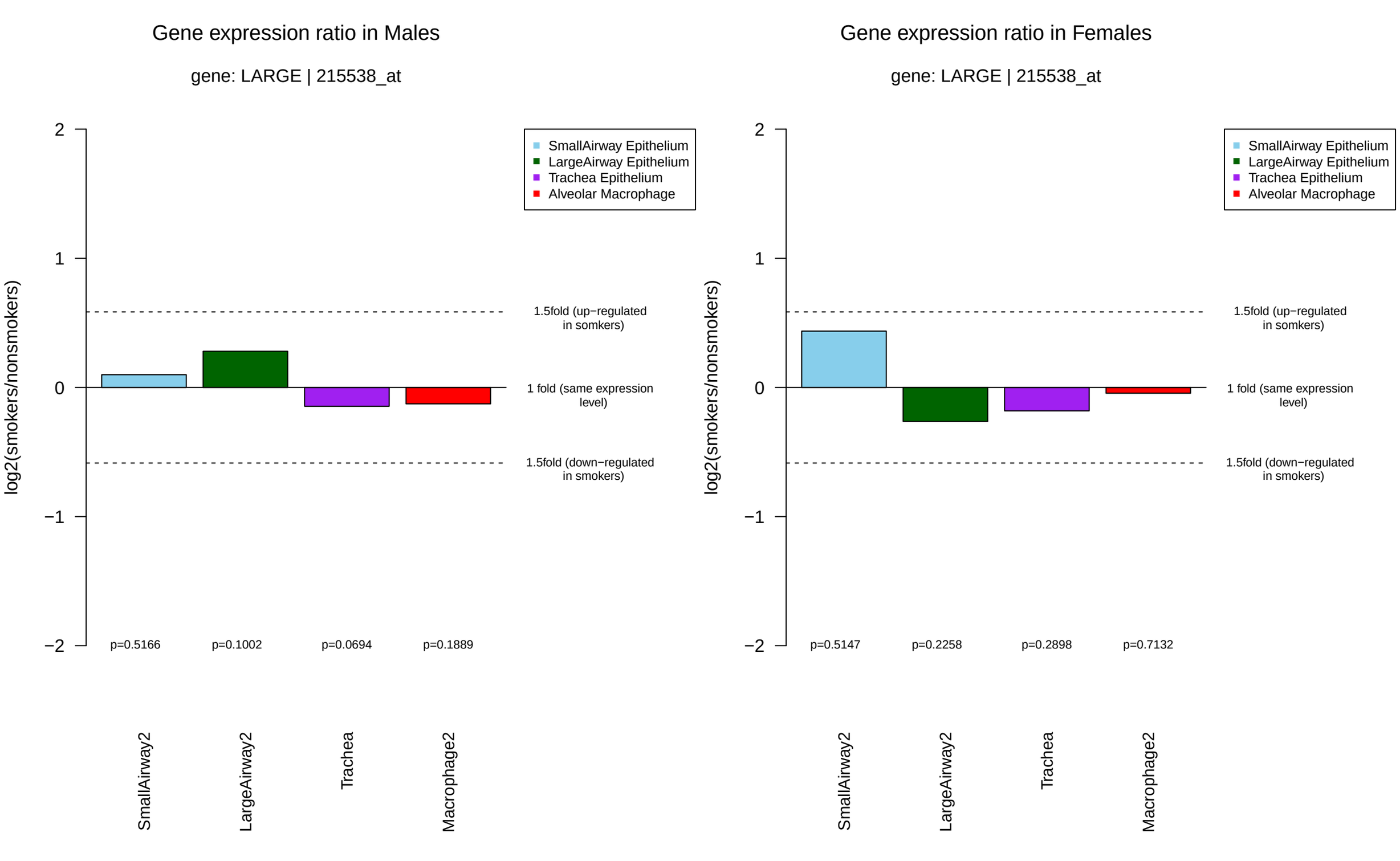
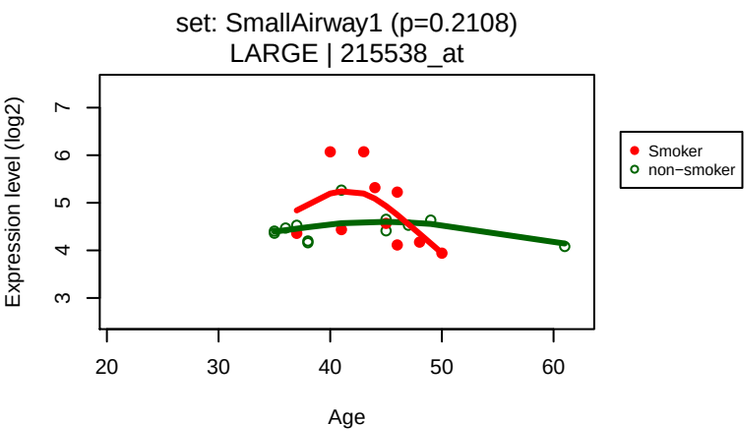
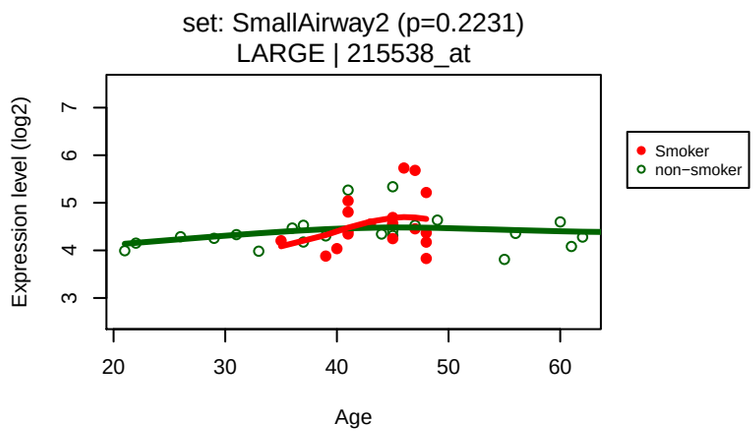
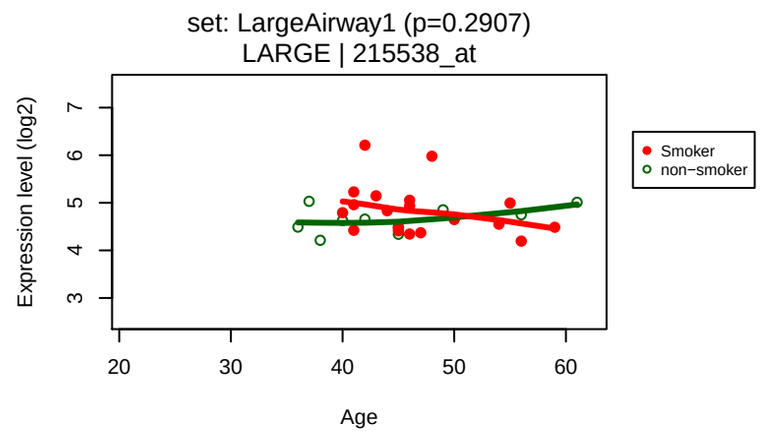

Gene expression of smokers and nonsmokers
gene: LARGE | 215538_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2108
p=0.2231
p=0.1798
p=0.2907
p=0.4953
p=0.0254
p=0.3082
p=0.1924
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

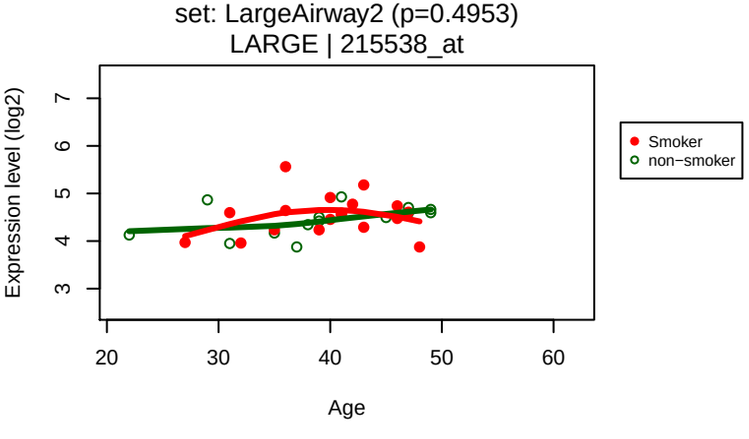

Gene expression ratio (smokers/nonsmokers)
gene: LARGE | 215538_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2108
p=0.2231
p=0.1798
p=0.2907
p=0.4953
p=0.0254
p=0.3082
p=0.1924
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

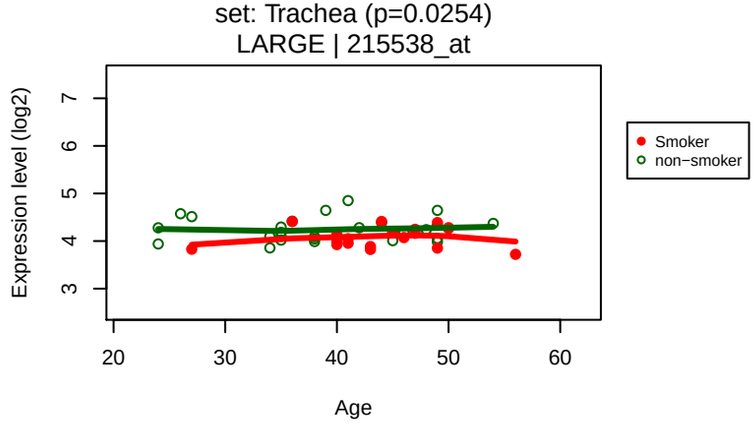

Gene expression ratio in Males
gene: LARGE | 215538_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5166
p=0.1002
p=0.0694
p=0.1889
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LARGE | 215538_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5147
p=0.2258
p=0.2898
p=0.7132
SmallAirway2
LargeAirway2
Trachea
Macrophage2

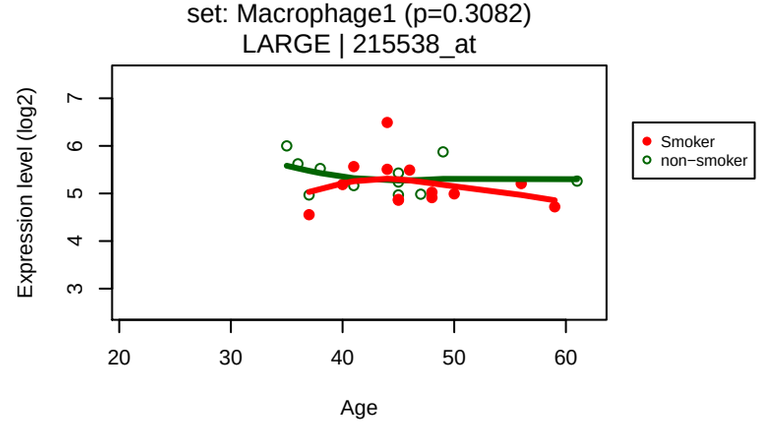

set: SmallAirway1 (p=0.2108)
LARGE | 215538_at
Expression level (log2)
Age
Smoker
non−smoker

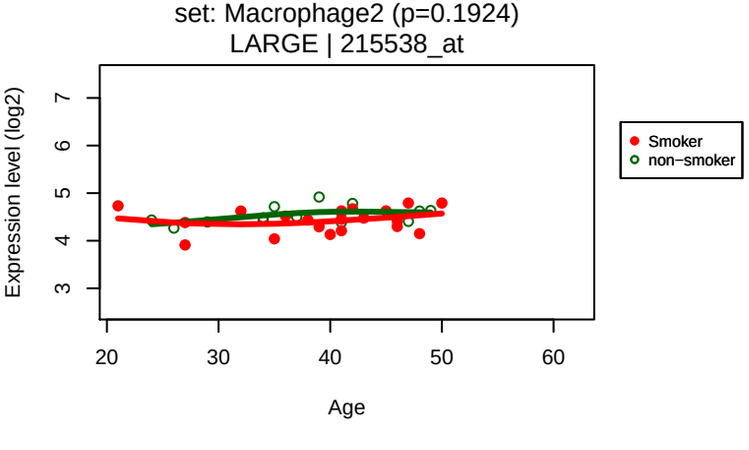

set: SmallAirway2 (p=0.2231)
LARGE | 215538_at
Expression level (log2)
Age
Smoker
non−smoker

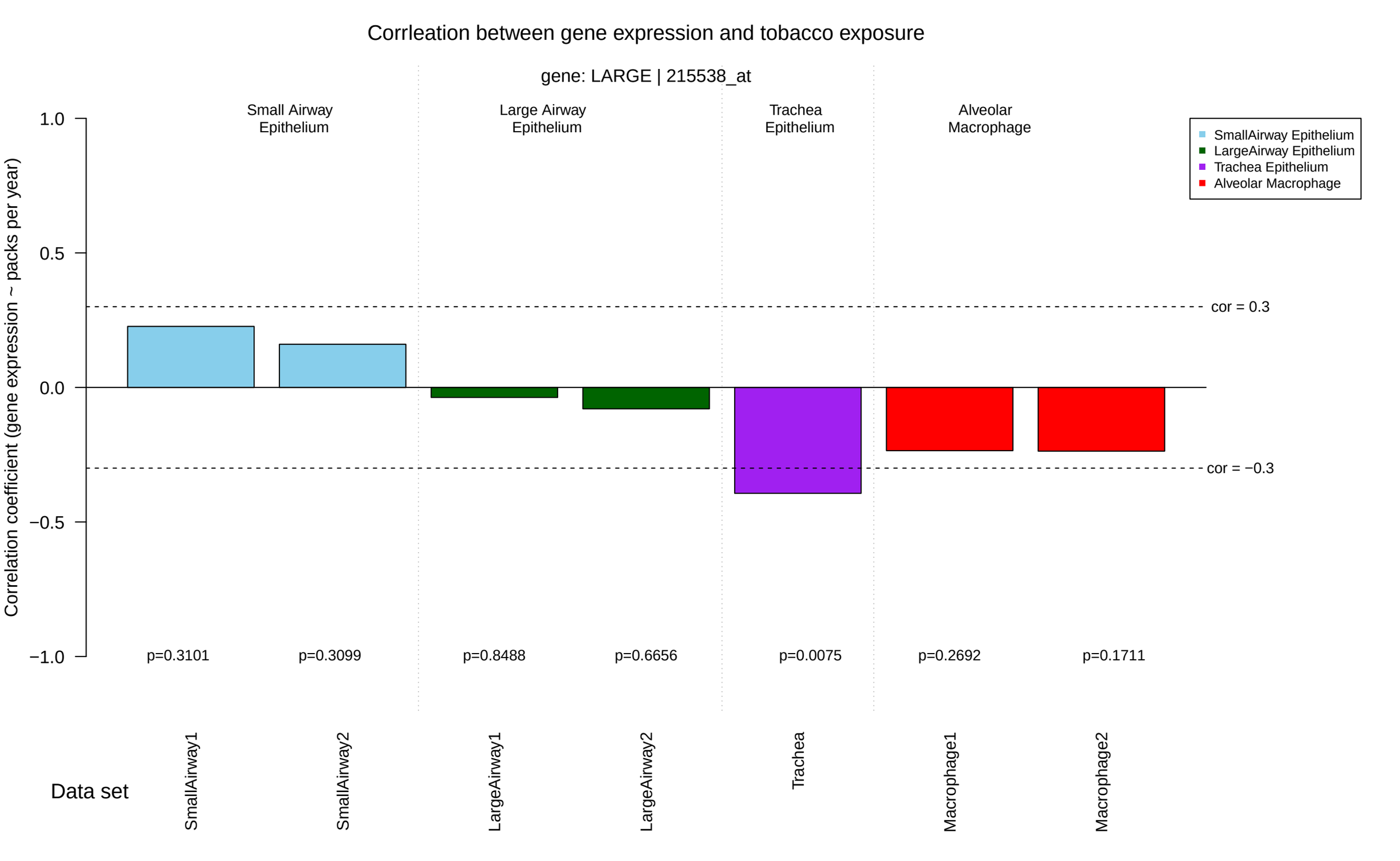

set: LargeAirway1 (p=0.2907)
LARGE | 215538_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4953)
LARGE | 215538_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0254)
LARGE | 215538_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3082)
LARGE | 215538_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1924)
LARGE | 215538_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LARGE | 215538_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3101
p=0.3099
p=0.8488
p=0.6656
p=0.0075
p=0.2692
p=0.1711
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2